

STATISTICS OF SEXUAL ISOLATION

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Sexual isolation is the tendency of individuals to avoid mating with those of another strain, race, or species. Isolation has been measured in *Drosophila* using a variety of methods and indices of isolation (e.g., Spiess, 1970; Parsons, 1973 for reviews). Statistical examination helps us evaluate the biological validity of these methods and indices.

The multiple choice method, a common experimental design, counts the instances of four types of matings when males and females of two races are placed together. This is diagrammed in Table 1. If the like-type (homogamic; cells A and D of the diagram) matings occur more frequently than expected by chance, the races are said to exhibit isolation. Statistics that are commonly used to measure such isolation are the Joint Isolation Index (Joint I) and Chi-square (Malogolowkin-Cohen et al., 1965; Zouros and d'Entremont, 1980). Two additional statistics which can measure isolation are Yule's V and Yule's Q (Bliss, 1967). These statistics differ in their properties (e.g., bias) and their utility for measuring isolation. How they differ and which are best depend on several factors.

Wattiaux (Wattiaux, 1964; Wattiaux and Elens, 1970; Elens et al., 1973) documented and discussed changes in measures of isolation over the time course of experiments (up to 120 min) with *Drosophila melanogaster*. The relative numbers of homogamic and heterogamic matings change; thus, measured isolation values depend on how long observations are continued. This may in part be due to changes with time in the sexual interactions in the pool of available mates, but part of this can be measurement bias due to sampling from the pool without replacement. Sampling without replacement occurs in most isolation experiments because, even if mated pairs are not physically removed, their copulation and subsequent refractory period for further mating biologically remove them from the pool of available mates. The ratios of available types, and the probability of matings, can change due to this removal, even though the presumed intrinsic isolation, or choice, probabilities remain unchanged. Manly (1980) discusses problems of estimating such intrinsic choice probabilities when sampling without replacement.

Another source of error in isolation estimates is the effect of vigor differences between races. Two hypothetically distinct properties that determine

whether matings occur are choice and vigor. We define *vigor* to be the likelihood that a male or female will engage in courtship and copulation, regardless of its partner. We define *choice* as the joint likelihood that a given male and female combination will mate, once they have initiated courtship. Sexual isolation occurs when choice is exercised between races, but vigor is usually considered distinct from isolation. In multiple choice experiments, vigor is measured from the marginal totals for a sex (F_1 vs. F_2 , and M_1 vs. M_2 in Table 1). Choice, or isolation, is measured from the interaction term ($A + D$ vs. $B + C$). Care should be exercised in relating these two population properties to underlying biological processes.

A final source of error considered here is the effect that unequal pools of mates may have on estimating intrinsic choice values. Unequal pools of mates are used in frequency-dependence experiments to establish whether scarcity or overabundance of a race will influence the intrinsic choice values.

We examine here the ability of four statistics to distinguish choice from vigor or unequal pools, and their sensitivity to sampling without replacement. The statistics, Joint I , Yule's V , Chi-square, and Yule's Q , are tested using simulated multiple choice experiments. They are compared under conditions of no intrinsic isolation, symmetrical isolation, asymmetrical isolation, vigor differences, and unequal pools of mates. The effects of sampling without replacement from a limited pool of mates are compared with asymptotic measures (i.e., sampling from an infinite pool).

MATERIALS AND METHODS

Simulated Matings.—Multiple choice experiments were simulated for cases in which the four parameters of choice, vigor, pool size, and percentage of mated individuals in the pool were varied. Each trial started with a pool of 80 male-female pairs, with fewer than or more than 40 pairs per race in the case of unequal pool sizes (p). Pairs were drawn and mated using vigor (v) and choice (c) probabilities as illustrated in Table 1. Probabilities were taken from the range [0.10-1.0] to reflect both mild and extreme cases. The trial continued until a percentage, from 25% to 90%, of the unmated pool was removed. Each trial was replicated 150 times, and several trials were run for each param-

TABLE 1. Diagrams of the multiple choice design and simulation parameters (1-3). In the design, A , B , C , and D are numbers of matings of each type, with female row (F_1 , F_2) and male column sums (M_1 , M_2), and total N matings. These letters correspond to observed numbers of matings used in the formulas for the isolation statistics (see text). Cases 1-3 diagram the relevant simulation parameters which differed between races. Choice (c) is the joint probability that a given pair will mate. Vigor (v) is the probability of drawing one or the other race for each sex. Pool size (p) is the number of individuals available for mating at the start of a test, for a total of 80 pairs.

Multiple choice design

		Male race		
		1	2	
Female race	1	A	B	F_1
	2	C	D	F_2
		M_1	M_2	N

1. Isolation

$$c_{11} = c_{22} = 1$$

$$c_{12} = c_{21} \leq 1$$

$$v = 1$$

$$p = 40$$

		Male race	
		1	2
Female race	1	c_{11}	c_{12}
	2	c_{21}	c_{22}

2. Vigor

$$c_{11} = c_{22} = 1$$

$$c_{12} = c_{21} \leq 1$$

$$v_{11} = v_{12} = 1$$

$$v_{21} = v_{22} < 1$$

$$p = 40$$

	Female		Male	
	1	2	1	2
Race:	v_{11}	v_{21}	v_{12}	v_{22}

3. Pool size

$$c_{11} = c_{22} = 1$$

$$c_{12} = c_{21} \leq 1$$

$$v = 1$$

$$p_{11} = p_{12} > 40$$

$$p_{21} = p_{22} < 40$$

$$p_{11} + p_{21} = 80, p_{12} + p_{22} = 80$$

	Female		Male	
	1	2	1	2
Race:	p_{11}	p_{21}	p_{12}	p_{22}

eter range. The simulation programs were written in Pascal and run on an Apple //e computer.

The probability of drawing a female of race 1 for courtship was determined by the product of its vigor and the frequency of race 1 females in the pool, relative to all females. The probability of drawing males of a particular race was determined similarly. Pairs drawn were mated with a choice probability; otherwise they were returned to the pool. For example, let $c_{11} = c_{22} = 1$, $c_{12} = c_{21} = 0.50$ be the choice probabilities (see Table 1.1), and $v_{11} = v_{12} = 1$, $v_{21} = v_{22} = 0.25$ be the vigor probabilities (Table 1.2), with pool sizes of $p_{11} = p_{12} = p_{21} = p_{22} = 40$ (Table 1.3). Thus, a female of race 1 is drawn with probability, $P = v_{11}/(v_{11} + v_{21}) = 1.0/(1.0 + 0.25) = 0.80$. A male of race 2 is drawn with $P = v_{22}/(v_{12} + v_{22}) = 0.25/(1.0 + 0.25) = 0.20$. They are then mated with a probability of 0.50. Asymmetrical isolation was simulated using $c_{11} = c_{22} = c_{21} = 1$ and $c_{12} < 1$.

Isolation Statistics.—Four indices were calculated from the number of matings for each experiment trial.

1) Joint Isolation Index (Malogolowkin-Cohen et al., 1965)

$$I = (A + D - B - C)/N$$

$$\text{var}(I) = (1 - I^2)/N$$

$$t(I) = I/\sqrt{\text{var}(I)}$$

2) Yule's V (Pielou, 1977)

$$V = (AD - BC)/\sqrt{F_1 F_2 M_1 M_2}$$

$$\text{var}(V) = (V)^2$$

$$\{-4/N + [AD(A + D) + BC(B + C)]/$$

$$(AD - BC)^2$$

$$- 0.75[(F_1 - F_2)^2/F_1/F_2/N$$

$$+ (M_1 - M_2)^2/M_1/M_2/N]$$

$$+ 0.50[AD - BC][F_1 - F_2][M_1 - M_2]/$$

$$F_1/F_2/M_1/M_2/N\}$$

$$t(V) = V/\sqrt{\text{var}(V)}$$

TABLE 2. Properties of isolation statistics for simulations of 1) symmetric and asymmetric isolation, 2) vigor effects with isolation, and 3) unequal pool sizes with isolation. Properties are for 50% sampling without replacement from a pool of 80 pairs. Simulation parameters for the three cases are defined in Table 1.

Statistic	Type I error	Power	Average bias	Average RMS
1. Isolation				
Joint I	0.033	1.044	-0.0013	0.0244
Yule's V	0.047	0.952	-0.0050	0.0208
Chi-square	0.053	0.988	-0.0060	0.0202
Yule's Q	0.065	0.544	-0.0247	0.0337
2. Vigor				
Joint I	0.139	1.884	0.0560	0.0810
Yule's V	0.051	1.990	-0.0088	0.0241
Chi-square	0.053	1.219	-0.0176	0.0298
Yule's Q	0.076	1.078	-0.0260	0.0406
3. Unequal pools				
Joint I	0.462	1.039	0.2703	0.1634
Yule's V	0.060	0.711	-0.0297	0.0465
Chi-square	0.053	0.949	-0.0697	0.0801
Yule's Q	0.105	0.441	-0.0544	0.0640

3) Chi-square (Bliss, 1967; Pielou, 1977)

$$X = 4(AD - BC)/N^2$$

$$\text{var}(X) = 16(F_1 F_2 M_1 M_2)/N^5$$

$$\chi^2(X) = X^2/\text{var}(X)$$

4) Yule's Q (Bliss, 1967; Pielou, 1977)

$$Q = (AD - BC)/(AD + BC)$$

$$\text{var}(Q) = (1 - Q^2)^2/4$$

$$\cdot (1/A + 1/B + 1/C + 1/D)$$

$$t(Q) = Q/\sqrt{\text{var}(Q)}$$

In the above formulas, A , B , C , and D are the numbers of four mating types; F_1 , F_2 , M_1 , and M_2 are the female row and male column sums; and N is the total of matings as indicated in Table 1.

All of these statistics take the range $[-1, 1]$ from complete heterogamic to complete homogamic matings. The variance of each statistic is given, and the t -statistic of each for calculating significance, except for number 3 above, for which the Chi-square is used for significance. This statistic is the usual contingency chi-square but has been normalized by $4/N$ to give the range $[-1, 1]$. Notice that the Joint

TABLE 3. Properties of Yule's V in relation to sampling without replacement (percentage mated). Simulation parameters for the three cases are listed in Table 1.

Percentage mated	Type I error	Power	Average bias	Average RMS
1. Isolation				
25	0.043	0.651	-0.0041	0.0314
50	0.047	0.952	-0.0050	0.0208
75	0.062	0.997	-0.0030	0.0158
90	0.075	0.994	-0.0042	0.0131
2. Vigor				
25	0.059	0.756	-0.0140	0.0328
50	0.051	1.990	-0.0088	0.0241
75	0.100	1.725	0.0179	0.0416
90	0.214	1.347	0.0514	0.0775
3. Unequal pools				
25	0.060	0.613	-0.0155	0.0636
50	0.060	0.711	-0.0297	0.0465
75	0.055	0.936	-0.0271	0.0405
90	0.032	1.951	-0.0273	0.0385

Isolation Index (number 1 above) depends on a sum of homo- and heterogamic matings, while the other three depend on the product of these matings.

Properties of the Statistics.—The relative utility of these statistics for detecting isolation is found by comparing their properties for hypothesis testing and estimation. Sample averages were calculated for each statistic and for its variance over the replicate trials. The number of significant statistics ($P < 0.05$) were tabulated using the probability level of the t or X^2 statistic. An asymptotic estimate for each statistic was calculated based on an infinite mating pool and the choice probabilities.

Two hypothesis testing properties of a statistic are its type I error and power. Type I error is the probability of falsely rejecting the null hypothesis of no isolation. This was measured by comparing the percentage of significant statistics over trials (measured alpha) where the asymptotic estimate was zero. Power is the probability of correctly rejecting the null hypothesis at some true level of isolation. Power was determined as the regression of the logit of measured alpha ($\ln \alpha - \ln[1 - \alpha]$) on the asymptotic estimate. Two estimation properties are bias and efficiency. Bias is a measure of the tendency of the sample estimate to deviate from the true (asymptotic) value and is calculated as the average deviation of sample estimates from the asymptotic values. Efficiency is a measure of the variance associated with the sample estimate. Here we report it as the root mean square deviation (RMS) of the sample from the asymptotic values, with smaller values indicating higher efficiency.

RESULTS AND DISCUSSION

The four statistics were compared for ability to accurately detect isolation when these conditions were varied: 1) symmetric and asymmetric isolation, 2) sexual vigor, 3) pool size or frequency dependence, and 4) percentage of matings drawn from the pool. Symmetric and asymmetric isolation trials yielded nearly identical results, so we present them combined as isolation. The properties of type I error, power, estimate bias, and efficiency are compared in Table 2 for cases 1–3. Type I error at or below 0.050 indicates a good hypothesis test, while large values of power are best. For estimation, smaller absolute bias and RMS are best.

Under all conditions, Yule's Q has relatively poor properties: its type I error and bias are large and its power is low. The other three statistics are similar for cases of isolation only (Table 2.1). Joint I is sensitive to vigor and pool size effects while the others are not (Table 2.2–2.3). This occurs because the calculation of Joint I does not remove the marginal effects (vigor or pool size) from the interaction (choice) effects. The other statistics do exclude mar-

ginal effects. Over all conditions, Yule's V and Chi-square provide the best hypothesis test and estimation. These two are nearly equivalent, as might be suspected from the similarity of their formulas. These results suggest that the commonly used Joint I should be avoided in future reports.

The effects of sampling without replacement are similar for all statistics and are shown for Yule's V in Table 3. Isolation, vigor, and unequal pool sizes yield somewhat different results here. Sampling less than 50% of the pool leads to a loss in efficiency (RMS is high) because fewer total matings are recorded. When vigor is a factor, nearly complete sampling (75–90%) can lead to high type I error, since the less vigorous individuals, all of one race, are the last ones to mate. In general, the results suggest that experiments collecting about 50% of the mating pool will yield the best estimates of isolation.

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